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A Low-cost IoT-based Meteorological System Using LoRaWAN and Embedded Technologies: Architecture and Future Trends

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Abstract

The field of meteorological station development is undergoing continuous advancements, driven by the pursuit of more precise data acquisition while also maintaining cost-effectiveness. To achieve this objective, governments and businesses are increasingly harnessing Internet of Things (IoT) platforms to deploy hyperlocal and highly sophisticated meteorological stations. These stations are designed to offer real-time analysis of weather conditions and forecasts with unparalleled precision. In this study, we have designed and implemented a robust yet affordable meteorological system capable of collecting various weather parameters. This system is integrated with a low-cost, long-range data transmission technology, enabling multiple nodes to access the internet through the LoRaWAN network server. Additionally, we have developed a user-friendly graphical user interface (GUI) application for visualizing meteorological data. Our proposed solution demonstrated impressive capabilities. The system consistently recorded and transmitted essential meteorological parameters, such as temperature, humidity, pressure, and wind speed, with high accuracy and reliability. Additionally, our GUI application facilitated user-friendly access to this data, offering clear visual representations of weather conditions and station performance.

Contribution of the Paper: The study enhances meteorological station development by introducing an affordable and robust system, leveraging IoT and LoRaWAN technology, to provide real-time and highly accurate weather data through a user-friendly graphical interface.

Keywords: Internet of Things, Meteorological Station, Long-Range Wide Area Network, Low-Power Wide Area Network

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1. Introduction

The world of environmental monitoring applications, including weather monitoring, smart cities, and smart agriculture, constantly expands with new technologies and solutions to acquire precise data readings utilising cost-effective approaches. A meteorological station is a facility on land

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or water that measures vital parameters for climatological studies to understand hydro-meteorological systems better and demonstrate patterns over time that might be connected to environmental change and accurate weather forecasting. Meteorological stations comprise machines, sensors, data storage, wireless communications, batteries, and solar collectors connected to the cloud via the Internet, cellular, or satellite communications for recording and transmitting meteorological parameter data, including air, wind speed, temperature, rainfall, atmospheric pressure, wind speed and direction, humidity, cloud height, light levels, and visibility [1].

The foundation of weather forecasts has always been based on weather observations and predictions, with different forms of meteorological models, including Global Models (for weather prediction across the globe) and Mesoscale Models (for smaller regions). The demand for microclimate weather is increasing for accurate local weather prediction since factors such as topography, irrigation, shelterbelts, etc. can affect local weather patterns. In light of this, it is necessary to establish a network of inexpensive weather stations and remote sensors [2]. Low-cost meteorological stations are weather monitoring systems that are designed to provide accurate weather data at an affordable price. These stations typically use cost-effective sensors and data collection methods to measure temperature, humidity, wind speed, wind direction, precipitation, and other meteorological variables. The data collected by these stations can be used for weather forecasting, climate monitoring, and various other applications. Some popular examples of low-cost meteorological stations include the Davis Instruments Vantage Pro2 [3] and the Ambient Weather WS-2902A [4].

The meteorological system is sought after by specific industries, such as agriculture and astronomy, to provide them with a guide on yield or work quality improvements and accurate and better weather forecasting [5]. Other application areas of the meteorological system include tourism and transportation, which can forecast and plan for the future when it comes to the local climate; farmers can know when to plant or harvest their crops; and we can evacuate regions if hurricanes or floods are expected.

There is a wide range of connectivity options when building meteorological systems, including WiFi, Bluetooth, Zigbee, SigFox, NB-IoT, and Long-Range Wide Area Network (LoRaWAN). Cano presents a low-cost wireless meteorological station based on LoRa technology [6] . The system also proposes wireless connectivity using the Bluetooth communication standard, providing a reliable, portable, and low-cost tool. However, there are several limitations of Bluetooth when compared to LoRa when building a low-cost meteorological system, including: range, power consumption, cost, and interference, etc [7]. The downfall of networking technologies in the development of low-cost IoT-based meteorological systems is their high power consumption, which could potentially lead to problems when the system is only supported by battery power. A recent and attractive option is LoRa, an acronym for long-range; it operates power-efficiently on a radio frequency range. It is designed to run on low-power systems that require long-range connectivity while maintaining a low-cost status [8]. LoRa is a long-range communication technology that uses the free Industrial, Scientific, and Medical (ISM) radio bands available worldwide and can establish connections up to 10 km away. Because of its low power consumption, the battery can last up to 10 years. Meanwhile, it is also capable of sustaining noise levels of under 20 dB during demodulation [9]. These standards make LoRa a dependable IoT networking technology eventhough it can experience packet loss since wireless connections are often unstable [10].

LoRaWAN, also known as Long Range Wide Area Network, is a wireless communication technology developed explicitly for facilitating connectivity inside the Internet of Things (IoT) ecosystem [11]. This technology facilitates energy-efficient and extensive networking for Internet of Things (IoT) devices. This technology can send data over considerable distances, typically spanning multiple kilometers, rendering it well-suited for applications necessitating remote monitoring or control. LoRaWAN devices exhibit energy efficiency, enabling them to function for prolonged durations while relying solely on battery power. This technology facilitates bidirectional communication, enabling devices to transmit data to and receive instructions from a centralised network. The integration of security measures in LoRaWAN ensures the safeguarding of data integrity and privacy. Additionally, the scalability of LoRaWAN enables its applicability in networks of many magnitudes, ranging from small-scale deployments with a limited number of devices to extensive implementations. This technology has been utilised in various domains, encompassing agriculture, smart cities, industrial automation, and environmental monitoring.

In comparison to more expensive meteorological stations, low-cost ones have a number of benefits, including:

- Affordability: The most significant advantage of low-cost meteorological stations is their affordability. These are far less expensive than high-end weather stations, making them more accessible to a larger variety of consumers.
- Easy to install and use: Low-cost meteorological stations are usually designed to be easy to install and use. They often come with simple instructions and user-friendly interfaces, making them accessible to people with little to no technical background.
- Portability: Low-cost weather stations are small and light, which makes it easy to move them from one place to another.
- Wireless connectivity: low-cost meteorological stations come with wireless connectivity options, making it easy to access weather data from anywhere.

- Wide range of applications: Low-cost meteorological stations can be used for a wide range of applications, including personal weather monitoring, agriculture, energy production, and more.

Subsequently, our contributions in this paper are as follows:

1. We developed a stable low-cost meteorological system capable of recording weather parameters. The developed system comprises MCUs, communication units, and sensors that record temperature, humidity, pressure, wind speed, and air quality.
2. We powered the developed meteorological system with a low-cost and long-range data transfer technology to provide multiple nodes access to the internet through the LoRaWAN network server - The Things Network (TTN).
3. We utilised ThingSpeak IoT analytics to collect, and organise the sensors' data and apply algorithms to make sense of it [12].
4. We developed a Graphical User Interface (GUI) for displaying meteorological data in the form of graphs and charts. Through visually intuitive representations, such as line graphs, bar charts, and heatmaps, users can swiftly grasp meteorological trends, fluctuations, and anomalies.

Our proposed solution is a meteorological station that is low-powered, portable, lightweight, and easy-to-deploy. In comparison to other wireless communication technologies, such as WiFi and cellular networks, LoRaWAN's low power consumption is one of the most notable characteristics of our proposed system, which uses spread-spectrum modulation to transmit data over long distances while consuming minimal energy.

The remaining sections of this work are organised as follows. The relevant literature is reviewed in Section II. Section III presented the components and design of the system. In Section IV, we discussed system testing. In Section V, we discussed future trends, and the conclusion was presented in Section VI.

2. Related Work

The meteorological system uses weather sensors, power supplies, data collectors, and cloud platforms to measure, transmit, record, and analyse weather data such as wind, rain, snow, sun radiation, and rain. Weather is recorded and analysed to produce accurate results and forecasts. Benefiting multiple industries by providing the current, past, and future meteorological conditions, allowing them to prepare and adapt. One of the most affected industries by micro-scale meteorology systems is agriculture, where balance must be maintained between variables like temperature or soil moisture to produce the optimum crop yield [13]. Similarly, astronomers rely on sky transparency forecasts to select suitable times for observation.

Recent advances in sensor technology have made it possible to create low-cost meteorological systems that provide high-quality weather data. For example, low-cost temperature and humidity sensors such as BME280 or DHT-11/22 are now widely available and can provide accurate and reliable data. Similarly, low-cost drone-based air quality sensors are used to measure wind speed and direction with a high degree of accuracy [14]. The low-energy Arduino MCU based on an Atmel 8-bit AVR microcontroller is capable of operating as the base of a meteorological station, which commands the multiple connected sensors [15]. Sensors are efficient devices used for data collection by monitoring physical conditions and are often installed on modules [7].

Rainfall gauges are great cost-effective tools for rain data observation and recording, or just functioning to provide secondary results. Certain users who may not have access to reliable satellite imagery, or just prefer to acquire more accurate local data, utilise rainfall gauges in their meteorological systems. Research indicates that rain gauges are the most common devices used to accurately measure the rainfall rate in specific locations [16]. A tipping bucket with a sensor is another local and digital option for rain data collection. The bucket fills up with rainwater and eventually tips, while the sensor captures the bucket tip timings.

2.1. LPWAN

The focus on cost mitigation in IoT meteorological stations must be maintained without major losses in networking performance. This has influenced the adaptation of Low-Power Wide Area Network (LPWAN) solutions, which offer communication using sub-GHz. Sub-GHz refers to communication technology that operates on the unlicensed ISM bands under 1 GHz (e.g., 315, 433, 902, and 928 MHz); these are frequently used in the fields of science, medicine, and manufacturing. Naturally, these channels provide increased network area coverage due to their lower frequencies compared to the wireless standard of 2.4 GHz. The demand for the development of IoT systems that feature low cost, low power, and long range connectivity, influence the implementation of lower ISM band devices such as Lora, SigFox, and BLE [17].

2.2. LoRaWAN

With the help of a gateway, LoRaWAN changes the normal LoRa mesh network into a star network. This network architecture allows for the connection of an extended number of nodes while providing sophisticated authorisation [18]. The LoRa protocol adapts all of the previously stated sub-GHz features. It functions on the unlicensed sub-GHz ISM bands, which are different depending on the geographical location, for example LoRaWAN in Europe or USA operates on the frequencies of 868 MHz and 915 MHz. Depending on the quality of the antenna, the low-frequency

technology offers a range of 10 km or more while maintaining customisable data rates of up to 300 kbps [19]. Additionally, LoRaWAN's demodulation and multiple channels offer smooth scaling of the system in the future. TTN and ThingSpeak analytics provide further security and organisation [12].

LoRaWAN packets are AES128 encrypted and a frame counter is applied to further boost network security from attack attempts. TTN adds another layer of security with its application's encryption method. The application utilises three unique keys, AppSKey, DevAddr, and NwksKey, creating an additional security layer, which ensures that only authorised devices can communicate with the server. The AppSKey is used to encrypt the whole payload, including the frame counter. Whereas the DevAddr is signed by the NwkSkey at the end of the node, this provides verification of integrity when validated by the network's gateway [20].

Based on our research of the aforementioned literature, we presented a user-friendly meteorological system. We created an additional GUI application for data visualisation in meteorological systems to present atmospheric data in a user-friendly format for ease of use, rapid decision-making, and improved communication, as opposed to the unorganised and ambiguous presentation typical of other solutions in the literature.

3. Systems Components and Design

A weather station must have certain build qualities to ensure that all of the targeted data can be collected by the sensors. Since the meteorological station is mounted outside, it must be suitable for all weather conditions. This can be achieved with an outdoor electronics enclosure consisting of a see-through plastic front, which ensures that light intensity data is collected by the sensors inside. Regarding networking, a LoRaWAN gateway is deployed, and the weather nodes are equipped with LoRa. This enables communication between end nodes and the gateway while taking advantage of a star-shaped network topology. Collected data is forwarded to the TTN, which integrates ThingSpeak, from which ThingSpeak's API is utilised in a web application to obtain real-time data updates.

A system design is illustrated in Figure 1, using LoRaWAN the end devices (meteorological sensors) transmit payloads directly the LoRAWAN gateways within the range. As illustrated in the figure, the system uses sensors to measure temperature, humidity, wind speed, wind direction, and other meteorological variables and transmits the data wirelessly to a central location using the LoRaWAN protocol. A LoRaWAN gateway receives the messages from the meteorological stations and forwards the data to the LoRaWAN network server (the Things network). This makes it possible to monitor weather conditions remotely without needing a physical connection to the station. The LoRaWAN protocol provides long-range, low-power communication, making it ideal for use in remote or rural locations where a physical connection to the station may not be possible, and make it possible to deploy multiple stations in a single location, providing a more comprehensive view of local weather conditions. From the Things Network server, the sensor data were then pooled to the ThingSpeak server, which offers several advantages for centralised storage and visualisation including scalability, flexibility, and real-time analysis. As can be seen in Figure 1, ThingSpeak provides real-time analytics capabilities using MATLAB functions that allow us to analyse our data as it is being collected with sensors [21]. This can be useful for detecting trends or anomalies in your data and for triggering alerts or notifications based on specific conditions or the application of machine learning algorithms on the data.

3.1. Meteorological Sensors

Meteorological sensors are instruments used to measure various weather parameters such as temperature, humidity, atmospheric pressure, wind speed and direction, precipitation, solar radiation, and more. These sensors are used in a wide range of applications including weather forecasting, climate research, agriculture, transportation, aviation, and renewable energy.

Given that the DHT-11/22 sensor provides real-time temperature and relative humidity readings. Since relative humidity provides the ratio of the amount of moisture in the air at the time of measurement to the maximum amount that air at that temperature could hold, we calculated an approximation of the dew point temperature using the following equation: $T_d = T_o - (100 - R_H)/5$, where T_d is dew point temperature (in degrees Celsius), T is the observed temperature (in degrees Celsius), and R_H is relative humidity (in percent). This indicates the amount of water vapour needed to achieve saturation increases as the temperature increases.

Another relatively common alternative to the DHT temperature sensor is the BME280 sensor. The BME280 similarly to the DHT-11/22, performs multi-sensory functions; it can collect data on humidity and temperature and is also able to take atmospheric pressure readings [22]. The BME280 sensor provides extra data that enables more advanced weather predictions and calculations without requiring extra sensory equipment, and it is a relatively cost-effective sensor.

3.2. Wind Speed and Rainfall Rate Sensors

In a meteorological system, wind flow speed is a fundamental atmospheric quantity caused by air moving from high to low pressure, usually as a result of changes in temperature. The wind affects weather forecasting, maritime operations, construction projects, the metabolism rate of many plant species, and countless other implications. We used anemometers to record the wind speed, which were constructed in a fan-like shape to capture wind flow velocity. Usually, cups or other lightweight objects are attached

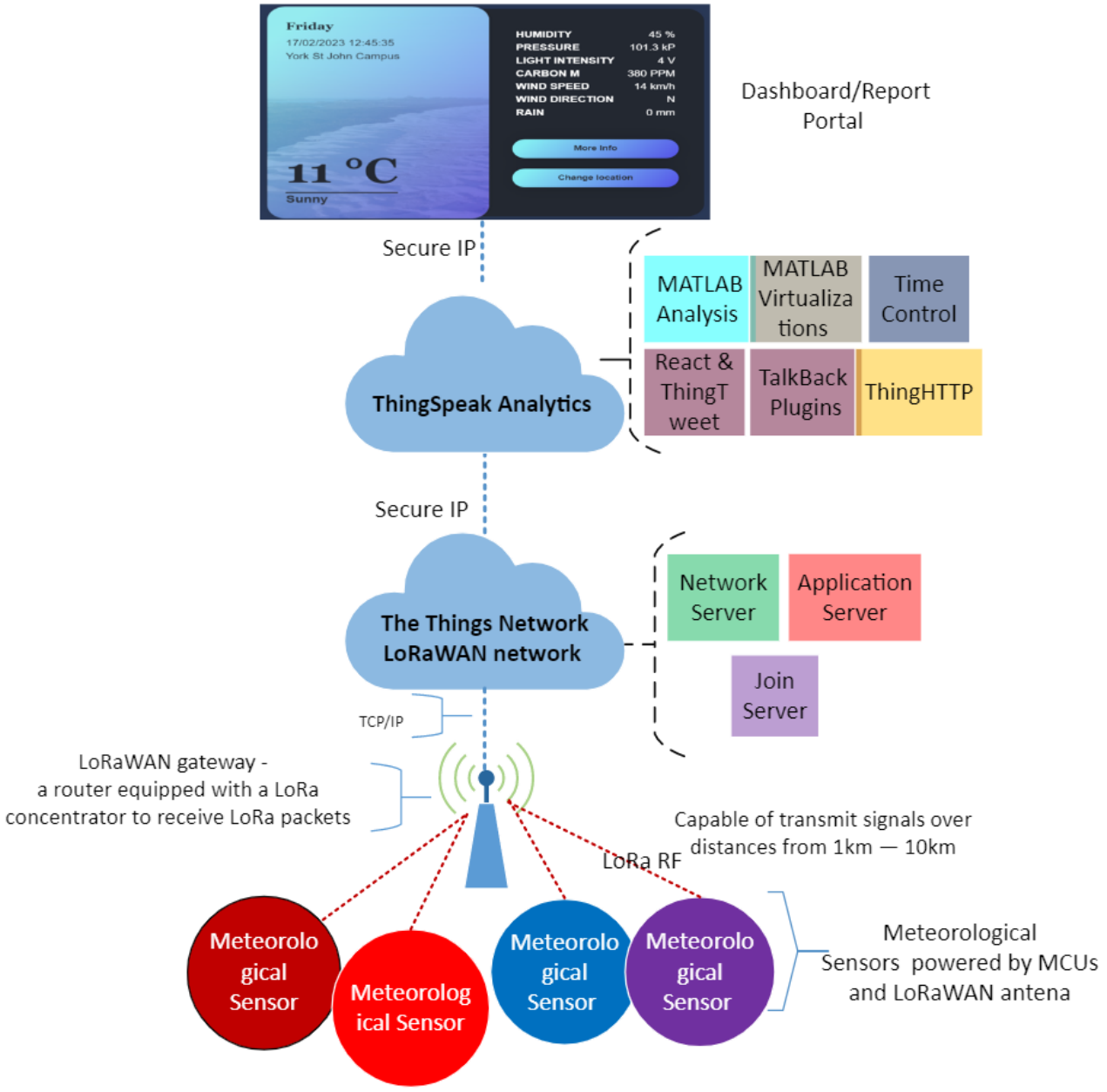


Figure 1: System Architecture of a Low-cost, LoraWAN-based Meteorological System

to the fan to increase the surface area of the fan. The increase in surface area on the blades provides more accurate readings on the wind speed and rainfall [23].

A rotor is placed onto a heavy steel base to ensure that the device stays static, and wings containing cups are installed onto the rotor. The cups are installed to ensure that all of the air from the wind velocity can be captured. Inside the device, a black and white pattern is embedded, with an optical sensor connected over it. While the device spins due to wind gusts, the optical sensor will decide whether the anemometer is in a black or white position. The voltage returned to the Arduino is dependant on the reflected shade. Ultimately, recording the changes between white and black can tell us the speed that the fan is spinning at; the more frequently the voltage switches, the faster the fan is spinning [24].

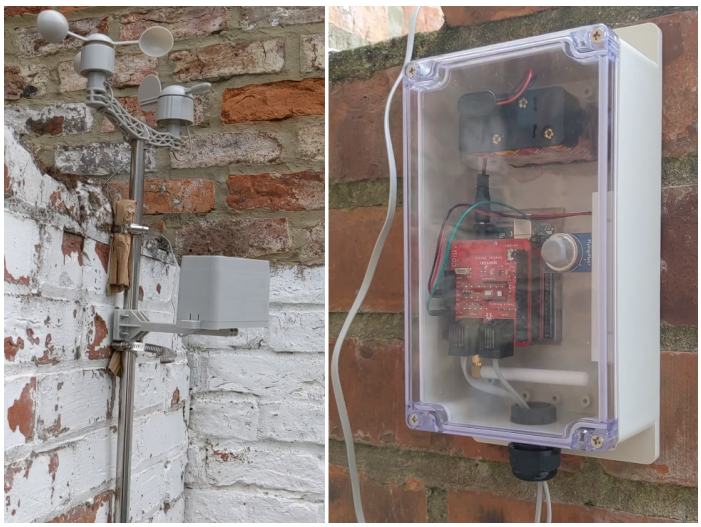


Figure 2: A Low-cost Meteorological System with Sensors and Anemometer

To collect air quality data, an MQ-135 sensor is installed using an analog pin on the Arduino MCU. It belongs to the MQ series sensor group, which consists of the most inexpensive and common gas sensors available. MQ-135 is dedicated to air quality measurements and is capable of collecting data on CO_2, CO, NH3, NOx, alcohol, benzene, and smoke levels in the air. In SETUP() the MQ-135 sensor is initialised and calibrated using the CALIBRATE() function. The readings are updated at every cycle with the UPDATE() function, and added to the MYDATA packet for TTN.

3.3. Systems' Electronics Components

We set up the Dragino LoRaWAN LPS8, an 8-channel gateway that enables communication between LoRa and the internet, specifically for the TTN application. To prevent exponential signal loss, we have installed the LoRa devices so that a line of sight is provided without the interception of buildings. This was required to maintain the stability of LoRa's long-distance communication. To enable LoRa communication, we install a Dragino LoRa shield onto the Arduino MCU. Each unique LoRa device receives an EUI and App Key. These two values, together with the application EUI are used for authorisation between the meteorological station and TTN.

3.3.1. The Things Network and ThingSpeak

The TTN IoT network is made up of a large number of gateways arranged in a star pattern, which are usually hosted by volunteers. This ensures a safe and long-term connection while also preserving the status of the data feed [25]. Devices containing sensors, like even the UK's flood network systems, utilise this technology to pass data to the internet. TTN is responsible for managing IoT systems on the internet side. With the Dragino LPS8 gateway registered with TTN application, it provides a live display of the captured LoRa data. Each meteorological station is separated into its own application, with the specific LoRa shield registered individually to each of the applications. Each station's data stream is sorted into its own application after reaching the gateway. To forward the data into a storage application, ThingSpeak is integrated. ThingSpeak's API makes the stored data retrievable by remote applications, capable of receiving near real-time sensor data. The communication method to the API is based on HTTP requests and authorisation keys (read/write). Data is stored in fields, and up to 8 of these fields are stored inside a channel. Read/write requests directly communicate with ThingSpeak's channels, to identify them apart, they are given a unique channel ID. Apart from fields, additional information like the description, latitude, longitude, and elevation is stored. Once a channel is setup, MATLAB processes and visualises the data. We applied ThingSpeak's MATLAB functions for data visualisation and computation. MATLAB is by default integrated with ThingSpeak, therefore it provides a default solution by applying standard algorithms to each field automatically. It is a preferred back-end environment by engineers due to its interactive system, which includes numeric computations, scientific visualisations and symbolic calculations, granting a comprehensive look into the gathered readings.

3.3.2. LoRa Technology

The meteorological station is equipped with Dragino LoRa shields, which can perform LoRa uplink and downlink. With the networking hardware assembled and installed, the programme includes instructions for establishing LoRa communication. The flowchart of the system can be seen in Figure 3.

3.4. Web Application

To present the data in a graphical form, we developed an application with a user interface to present the established data feed. The back end of the developed application was written in Flask (Python) to support scaling and provide the necessary security between the application and

ThingSpeak. The client side will have timed requests for the collection of data readings, coded in jQuery, allowing for weather data updates without the web page having to reload. Standard HTML, CSS, and Javascript were used to develop the front end. We deployed the web application by hosting it on a private server, which provided it accessibility to the internet. The flowchart of the web application can be seen in Figure 4.

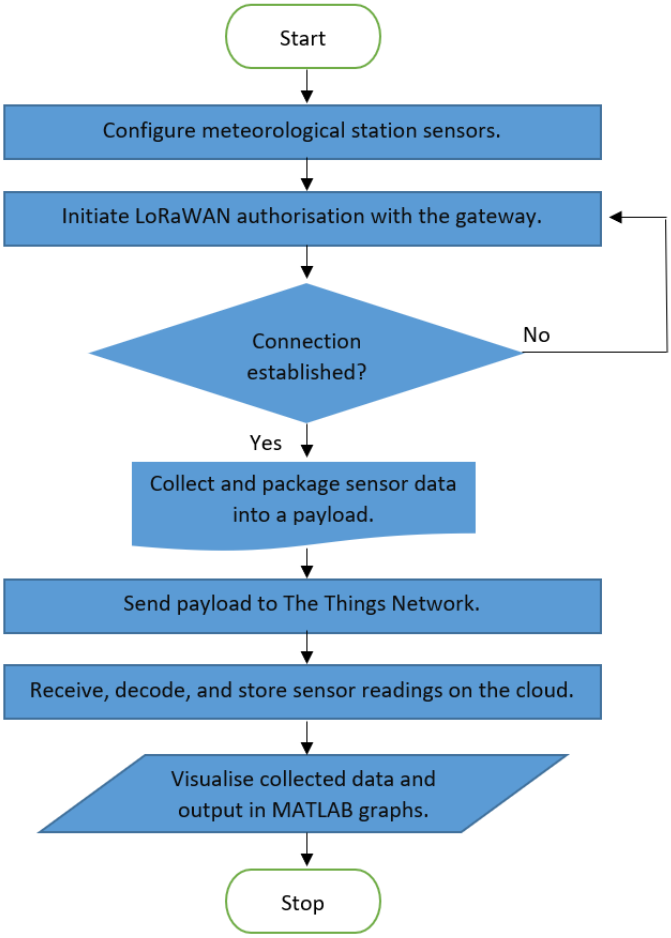


Figure 3: Meteorological System Flowchart

4. System Testing

General testing is performed, where the stations are launched online for a couple of hours to collect data. Testing continues until enough data is gathered for sufficient results, through which we are able to evaluate its complete life cycle. Beginning at the sensors, then the Arduino MCU, LPS8 Gateway, TTN, ThingSpeak, and finally ending its cycle at the web application.

Figure 5 presents the atmospheric data graph measured by a low-cost meteorological station, which depicts numerous atmospheric variables throughout time. The factors that are measured are temperature, humidity, pressure,

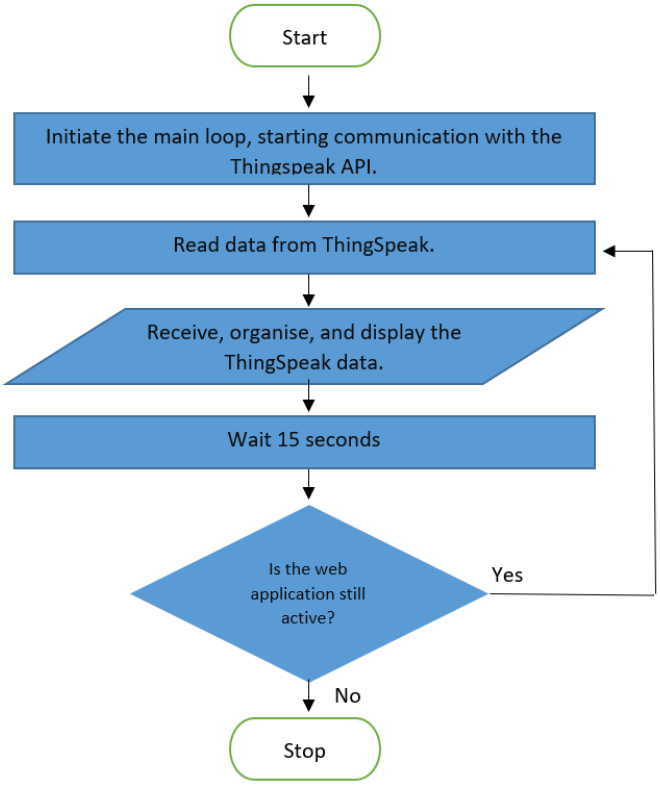


Figure 4: Web Application Flowchart

and light. The graph depicts the variable as a line with a time stamp on the x-axis and the variable's value on the y-axis. Each line was color-coded to indicate a distinct variable. The sensor-measured values and the atmospheric data can be seen in Figure 5, and the data was subsequently pushed to the web app designed to host the data. The developed GUI app displays information on the weather and stations on a mobile app to the users (in the form of numbers, charts, and graphs) as illustrated in Figure 6.

The first test involves the observation of active LEDs on the electronics and sensors when the Arduino MCU is connected to the batteries. This indicates if the meteorological station is receiving power. Thereafter, the MCU should begin calibrating the sensors and initialising a LoRa connection with the gateway. To inspect if a successful connection has been established, we open the TTN application and cheque if the status appears as "online". Each payload received by TTN was decoded and sent to ThingSpeak. Timestamps were evaluated to see whether each payload was registered.

A stable feed of readings was generated by both stations, and the LoRaWAN connection remained active during the testing phase. To obtain results on simultaneous station communication, metadata on the gateway used is extracted from TTN application payloads. The test begins by accessing the ThingSpeak channel and comparing the timestamps of the entries to TTN payload timestamps.

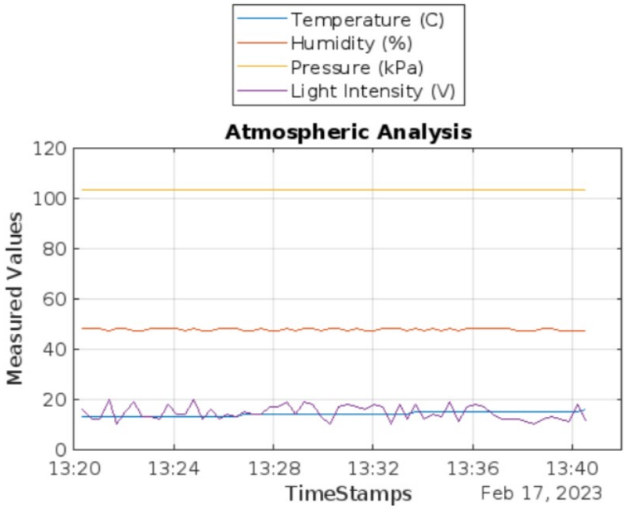


Figure 5: Atmospheric Data Measured by the Low-cost Meteorological Station

The results match which confirms that the correct data has been forwarded from the TTN to ThingSpeak.

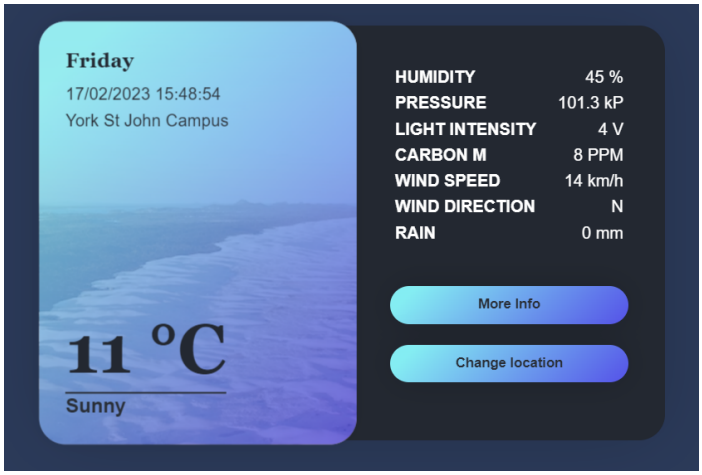


Figure 6: Application Testing

4.1. Web Application Functionality Test

The web application is launched in Google Chrome, and the user is directed to the main index page. Since both the time and the weather data change every 10 seconds, it can be assumed that the updates are going well. All of the expected results are displayed, including the temperature, humidity, pressure, light intensity, carbon monoxide, wind speed, wind direction, and rain.

5. Future Research Directions

The field of low-cost IoT-based meteorological systems that utilise LoRaWAN and embedded technologies offers numerous prospects for future research and opportunity. The primary objective is to improve sensor technology, focusing on both accuracy and cost-effectiveness. Scholars have the opportunity to investigate the advancement and incorporation of sophisticated meteorological sensors that possess the dual characteristics of cost-effectiveness and enhanced precision and resilience in the measurement of crucial variables including temperature, humidity, rainfall, wind speed, and air quality.

Energy harvesting systems represent a highly potential path for exploration and development. The exploration of techniques for harnessing energy from renewable sources such as solar power or kinetic energy has the potential to considerably prolong the functional duration of Internet of Things (IoT) devices. This would mitigate the necessity for frequent battery changes and enhance their sustainability, particularly in distant or hard-to-reach areas.

The utilisation of data processing and analysis will continue to play a pivotal role in forthcoming research endeavours. The investigation of data fusion techniques, which integrate data from diverse IoT devices and sources, has the potential to enhance the accuracy of weather forecasts by providing more extensive insights. The application of advanced data analytics and machine learning algorithms has the potential to extract significant patterns and trends from this dataset, hence augmenting the precision of meteorological forecasts.

The optimisation of LoRaWAN networks plays a crucial role in ensuring the smooth operation of meteorological Internet of Things (IoT) systems. Scholars have the opportunity to explore ways to enhance the scalability, dependability, and coverage of networks in order to maintain uninterrupted data transmission across vast geographical regions. The preservation of data security and privacy is of equal importance when it comes to transmitting information through these networks, particularly in the context of crucial meteorological data that has the potential to affect public safety and infrastructure.

Future research could potentially place greater emphasis on the development of real-time weather prediction models that rely on data gathered from Internet of Things (IoT) meteorological systems. These models have the potential to offer geographically specific weather forecasts, advance notice of natural disasters, and contribute to the understanding of climate trends. Furthermore, there is a continued emphasis on the advancement of economical hardware solutions, which includes the utilisation of low-power microcontrollers, communication modules, and inventive approaches for integrating sensors.

Moreover, it is imperative to investigate the feasibility of implementing these systems in many contexts, encompassing both urban and rural areas. Urban applications may primarily concentrate on the monitoring of microclimates and urban planning, whereas rural applications may specifically address agriculture and resource management, hence presenting distinct difficulties and opportunities.

The exploration of integrating meteorological data with other IoT systems is a promising opportunity for academics

to examine the potential synergies and benefits that can arise. Specifically, this integration has the potential to enhance and augment many IoT applications such as smart agriculture, smart cities, and disaster management. The exploration of citizen science programmes and community engagement tactics is recommended for the purpose of involving local communities in the deployment and maintenance of low-cost meteorological devices. This approach would serve to enhance data gathering efforts and foster community resilience.

The developed prototype represents a comprehensive IoT LPWAN (Low Power Wide Area Network) solution designed for the acquisition and analysis of meteorological data. Throughout the prototype development phase, several challenges emerged that were not initially apparent during the design stage. Notably, we had to carefully manage the memory capacity of the Arduino microcontroller when deploying multiple sensors on this specific MCU. Both software and hardware capabilities were stretched to their limits to accommodate the required number of sensors. Additionally, our cloud solution on Thingspeak had to be divided into multiple channels to handle the substantial volume of incoming data from the monitoring stations.

A critical consideration for future work pertains to the weatherproofing and outdoor installation of these meteorological stations. While we successfully addressed rain protection, we encountered issues related to condensation caused by varying outdoor weather conditions, such as humidity fluctuations and sunlight exposure, which eventually affected the electronic components' functionality. Further research can explore potential solutions, such as equipping the stations with appropriately designed casings tailored to their specific environments. It is noteworthy that the model necessitates access to sunlight inside the casing to ensure adequate light levels for data collection.

Although the battery usage enabled limited remote deployments, it could not sustain continuous operation due to rapid depletion caused by the numerous active sensors. Future endeavours should investigate strategies to maintain battery charge, including the adoption of technologies such as solar panels or wind/water turbines.

The proposed model offers an affordable and remote meteorological monitoring solution, providing smaller organizations and individuals with an alternative for localized data collection. This approach promises greater accuracy compared to relying on public resources, such as online weather data, which provide averaged information over larger regions. Furthermore, agriculture entities could benefit from such technologies, utilising it to gain increased crop yield by adapting to the local real-time conditions.

6. Conclusion

In this study, we proposed a reliable, low-cost IoT-based meteorological system to provide a cost-effective and efficient solution for collecting and transmitting meteorological data. MCUs and sensors record temperature, humidity, pressure, wind speed, and air quality in the created system. The designed system sends real-time meteorological data, allowing real-time decision-making based on collected data. The proposed system was designed to be energy efficient, lowering running expenses and increasing equipment lifespan. The system was outfitted with LoRaWAN networking technologies, allowing it to be used in rural and distant areas. The developed system, as part of this paper's contribution, has several advantages in the current situation, including being less expensive than traditional high-end meteorological systems, increasing accessibility for measurement and calibration processes, having more portable and easy-to-use features, and being more easily usable, making it ideal for use in the field or in remote locations-the developed system.

References

[1] J. Crawford, S. D. Chambers, A. G. Williams, Assessing the impact of synoptic weather systems on air quality in sydney using radon 222, Atmospheric Environment 295 (2023) 119537.

[2] A. Cecilia, G. Casasanta, I. Petenko, A. Conidi, S. Argentini, Measuring the urban heat island of rome through a dense weather station network and remote sensing imperviousness data, Urban Climate 47 (2023) 101355.

[3] L. Y. Heng, M. E. Rana, R. Abdulla, H. K. Tripathy, Application of iot and cloud computing in e-cultivation, design and development recommendations for a smart weather monitoring system., In 2022 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC) (2022) 1–9.

[4] D. Instruments, Vantage pro2., Hayward, California, United States.

[5] F. A. Almalki, B. O. Soufiene, S. H. Alsamhi, H. Sakli, A low-cost platform for environmental smart farming monitoring system based on iot and uavs, Sustainability 13 (11) (2021) 5908.

[6] E. M. Cano-Cruz, F. López-Orozco, Design and development of a low-cost and portable meteorological system: meteoblue, Advanced Science Letters 21 (1) (2015) 83–87.

[7] K. Krishnamurthi, S. Thapa, L. Kothari, A. Prakash, Arduino based weather monitoring system., International Journal Of Engineering Research And General Science. (2015) 3, 452–458.

[8] S. Devalal, A. Karthikeyan, Lora technology-an overview, 2018 Second International Conference On Electronics, Communication And Aerospace Technology (ICECA) (2018) 284–290.

[9] E. Murdyantoro, R. Setiawan, I. Rosyadi, A. W. Nugraha, H. Susilawati, Y. Ramadhani, Prototype weather station uses lora wireless connectivity infrastructure, in: Journal of Physics: Conference Series, Vol. 1367, IOP Publishing, 2019, p. 012089.

[10] M. Vaezi, A. Azari, S. R. Khosravirad, M. Shirvanimoghaddam, M. M. Azari, D. Chasaki, P. Popovski, Cellular, wide-area, and non-terrestrial iot: A survey on 5g advances and the road toward 6g, IEEE Communications Surveys And Tutorials 24 (2) (2022) 1117–1174.

[11] A. Lavric, V. Popa., A lorawan: Long range wide area networks study., 2017 International Conference on Electromechanical and Power Systems (SIELMEN), IEEE (2017) 417–420.

[12] M. Maureira, D. Oldenhof, L. Teernstra, Thingspeak–an api and web service for the internet of things, World Wide Web, 25 (2011) 1–4.

[13] C. Muller, L. Chapman, C. Grimmond, D. Young, X. Cai, Sensors and the city: a review of urban meteorological networks., International Journal Of Climatology. (2013) 33, 1585–1600.

[14] N. Afshar-Mohajer, C.-Y. Wu, Use of a drone-based sensor as a field-ready technique for short-term concentration mapping of air pollutants: A modeling study, Atmospheric Environment 294 (2023) 119476.

[15] L. Louis, Working principle of arduino and using it as a tool for study., International Journal of control Automation, Communications and Systems.
[16] M. J. Alam, S. A. Rafi, A. A. Badhan, M. N. Islam, S. I. Shuvo, A. M. Saleque, Design and implementation of a low-cost iot based tipping bucket rain gauge, in: 2021 International Conference on Emerging Smart Computing and Informatics (ESCI), IEEE, 2021, pp. 279–283.
[17] G. Raj, A. Verma, P. Dalal, A. K. Shukla, P. Garg, Performance comparison of several lpwan technologies for energy constrained iot network, International Journal of Intelligent Systems and Applications in Engineering 11 (1) (2023) 150–158.
[18] K. Ioannou, D. Karampatzakis, P. Amanatidis, V. Aggelopoulos, I. Karmiris, Low-cost automatic weather stations in the internet of things, Information 12 (4) (2021) 146.
[19] A. Di Serio, J. Buckley, J. Barton, R. Newberry, M. Rodencal, G. Dunlop, B. O'Flynn, Potential of sub-ghz wireless for future iot wearables and design of compact 915 mhz antenna, Sensors 18 (1) (2018) 22.
[20] N. Blenn, F. Kuipers, Lorawan in the wild: Measurements from the things network., ArXiv Preprint ArXiv:1706.03086.
[21] S. Pasha, Thingspeak based sensing and monitoring system for iot with matlab analysis., International Journal Of New Technology And Research (IJNTR) (2016) 2, 19–23.
[22] P. D. P. Adi, A. Kitagawa, A performance of radio frequency and signal strength of lora with bme280 sensor, Telkomnika 18 (2) (2020) 649–660.
[23] K. Warnakulasooriya, Y. P. Jayasuriya, B. Sudantha, Generic iot framework for environmental sensing researches: Portable iot enabled weather station, in: 2018 International Conference on System Science and Engineering (ICSSE), IEEE, 2018, pp. 1–5.
[24] A. Shaout, Y. Li, M. Zhou, S. Awad, Low cost embedded weather station with intelligent system., 2014 10th International Computer Engineering Conference (ICENCO) (2014) 100–106.
[25] P. Barro, M. Zennaro, E. Pietrosemoli, Tltn–the local things network: on the design of a lorawan gateway with autonomous servers for disconnected communities, 2019 Wireless Days (WD) (2019) 1–4.